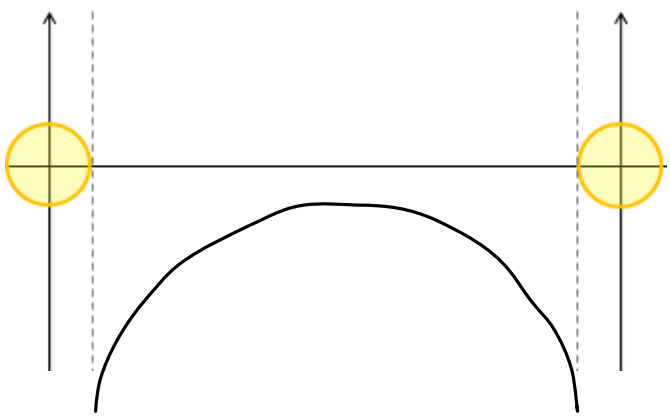


2	(a)	The gravitational potential at a point is the work done per unit mass by an external agent in bringing a point mass from infinity to that point.	
	(b)	<i>Gravitational potential at the centre of mass O</i> $= \left(-\frac{Gm}{0.5d}\right) + \left(-\frac{Gm}{0.5d}\right)$ $= -\frac{4Gm}{d}$ $= -\frac{4(6.67 \times 10^{-11})(3.5 \times 10^{30})}{2.0 \times 10^{11}}$ $= -4.67 \times 10^9 \text{ J kg}^{-1}$ Where d is the separation between the two stars.	
	(c)	By the principle of conservation of energy, Loss in kinetic energy of the asteroid = Gain in gravitational potential energy of the asteroid as it completely leaves the gravitational pull of the binary star system $\frac{1}{2}mv^2 - 0 = 0 - \left(-\frac{4GMm}{d}\right)$ $v = \sqrt{\frac{8GM}{d}}$ $= \sqrt{\frac{8(6.67 \times 10^{-11})(3.5 \times 10^{30})}{2.0 \times 10^{11}}}$ $v = 9.66 \times 10^4 \text{ m s}^{-1}$	
	(d)	(i)	Gravitational Potential / J Kg⁻¹ Gravitational Potential / J Kg⁻¹

				
		(ii)	The gravitational potential energy of the object will decrease as it moves towards star M. As the object leaves the centre of mass, the gravitational potential decreases at an increasing rate and since the mass of the object is constant, the gravitational potential energy of the object decreases.	

2	(c)	is a value that is numerically equal to the thermal energy transferred when unit mass of a	
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